



UITS

Future will be better than the past

University of Information Technology & Sciences

An initiative of *PHP Family*

Compiler Lab

Lab Report-5

Course Title : Compiler Lab

Course Code : CSE0613322

Submitted to:

Teacher's Name	Designation
Sonia Afroz	Lecturer
Tasneen Tanvir	Lecturer

Submitted by:

Student Name : Gaus Saraf Murady

Student ID : 0432410005101088

Semester : Autumn 2026

Batch : 55

Department : CSE

Date of Submission: 05.09.26

Experiment No. : 5

Experiment Name: Pattern Matching Problems

Given Test Cases :

Input	Output
$(ab)^*$	ab, abab, abab.....
ab^+	ab, abb, abbb.....
a^+b	ab, aab, aaa....b
$a^?b^+$	b, ab, ab, abbb

Code:

```
#include <bits/stdc++.h>
using namespace std;

int main()
{
    string n;
    cin >> n;
    cout << endl << endl;

    int ns = n.size();
    string temp;

    // (ab)*
    if(n[0] == '(' && n[ns-2] == ')') && n[ns-1] == '*')
    {
        cout << "[ (ab)* ]" << endl << endl;
        cout << "epsilon , ";
        temp += n[1];
        temp += n[2];
        for(int i = 0; i<3; i++)
        {
            cout << temp << " , ";
            temp += temp;
        }
        cout << " ... ";
    }

    // a?b+

    else if(n[1] == '?' && n[ns-1] == '+')
    {
```

```

        cout << "[ a?b+ ]" << endl << endl;
        cout << n[2] << " , ";
        temp += n[ns-2];
        for(int i = 0; i<3; i++)
        {
            cout << n[0] << temp << " , ";
            temp += n[ns-2];
        }
        cout << " ... ";

    }

    //ab+

    else if(n[ns-1] == '+')
    {
        cout << "[ ab+ ]" << endl << endl;
        temp += n[0];
        temp += n[1];
        for(int i = 0; i<3; i++)
        {
            cout << temp << " , ";
            temp += n[1];
        }
        cout << " ... ";

    }

    // a+b

    else if(n[1] == '+')
    {
        cout << "[ a+b ]" << endl << endl;
        temp += n[0];
        for(int i = 0; i<3; i++)
        {
            cout << temp << n[2] << " , ";
            temp += temp;
        }
        cout << " ... ";

    }

    cout << endl << endl;

    return 0;
}

```

Input	Output
$(ab)^*$	<pre>[(ab)*] epsilon , ab , abab , abababab , ...</pre>
ab^+	<pre>[ab+] ab , abb , abbb , ...</pre>
a^+b	<pre>[a+b] ab , aab , aaaab , ...</pre>
$a?b^+$	<pre>[a?b+] b , ab , abb , abbb , ...</pre>

Discussion:

Through this assignment, I learned how to identify and show output by the corresponding input string. Here are 2 of my observations and learnings:

- I particularly struggled with the case `a?b+` since it contained the ending `+` which overlapped with `ab+` case, showing the wrong result when input was given `a?b+`. The fix to which was to put the `else if` block for `a?b+` before the `else if` block for `ab+`. Hence when input matched both `n[1] == '?' && n[ns-1] == '+'` and not just ending `n[ns-1] == '+'`, the code would correctly get into the `else if` blocks for both `ab+` and `a?b+` blocks.
- I used strings as character arrays `string n; cin >> n;` which helped me control the pattern matching using for loops.

Conclusion:

I was able to complete the assignment successfully.